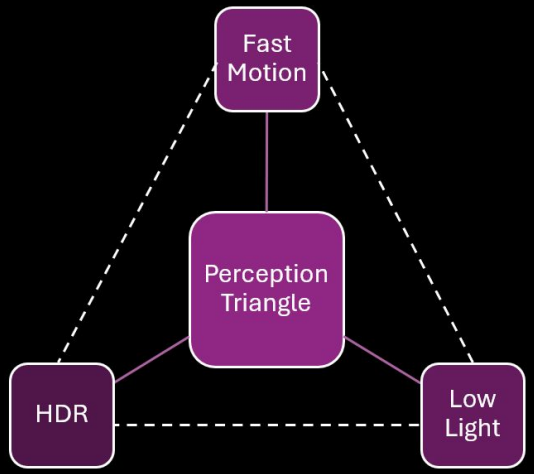
CS766 Final Presentation

Fast, Vectorized Color QBP + Generative Modeling for SPAD Imaging

Aryan Garg & Avery Gump

Problem Statement



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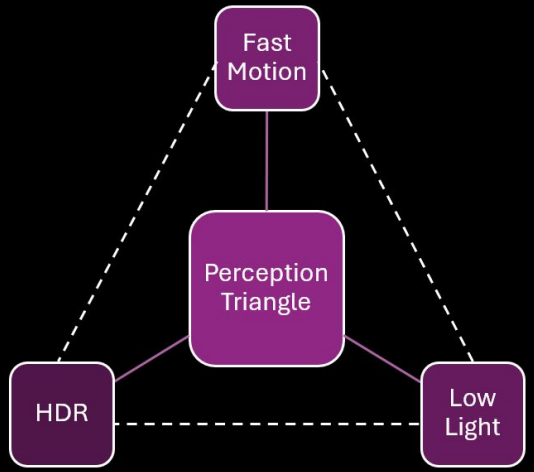

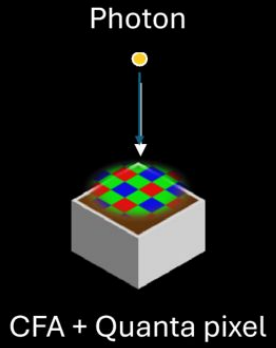

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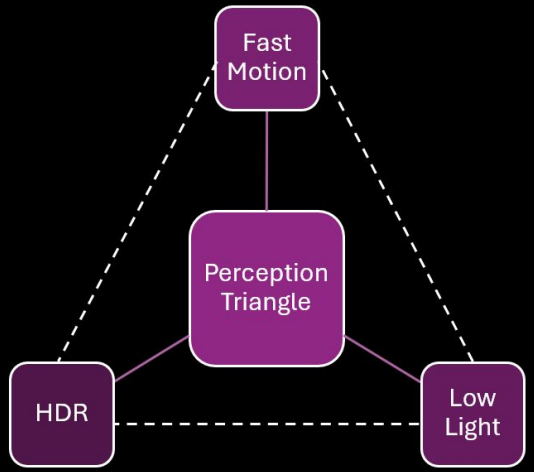
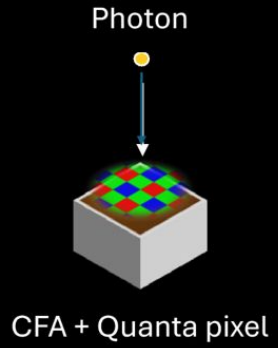




Our goal is to recover *high fidelity, color HDR images* from Single Photon Cameras.

Why is this problem important?

Burst photography with SPADs allows for high fidelity, high motion HDR imaging under different illumination levels, even at night time conditions!

Methodology

1. Reimplementing a classical burst SPAD ISP in PyTorch (core contribution)
   a. Vectorized reimplementation
   b. Much faster, same fidelity

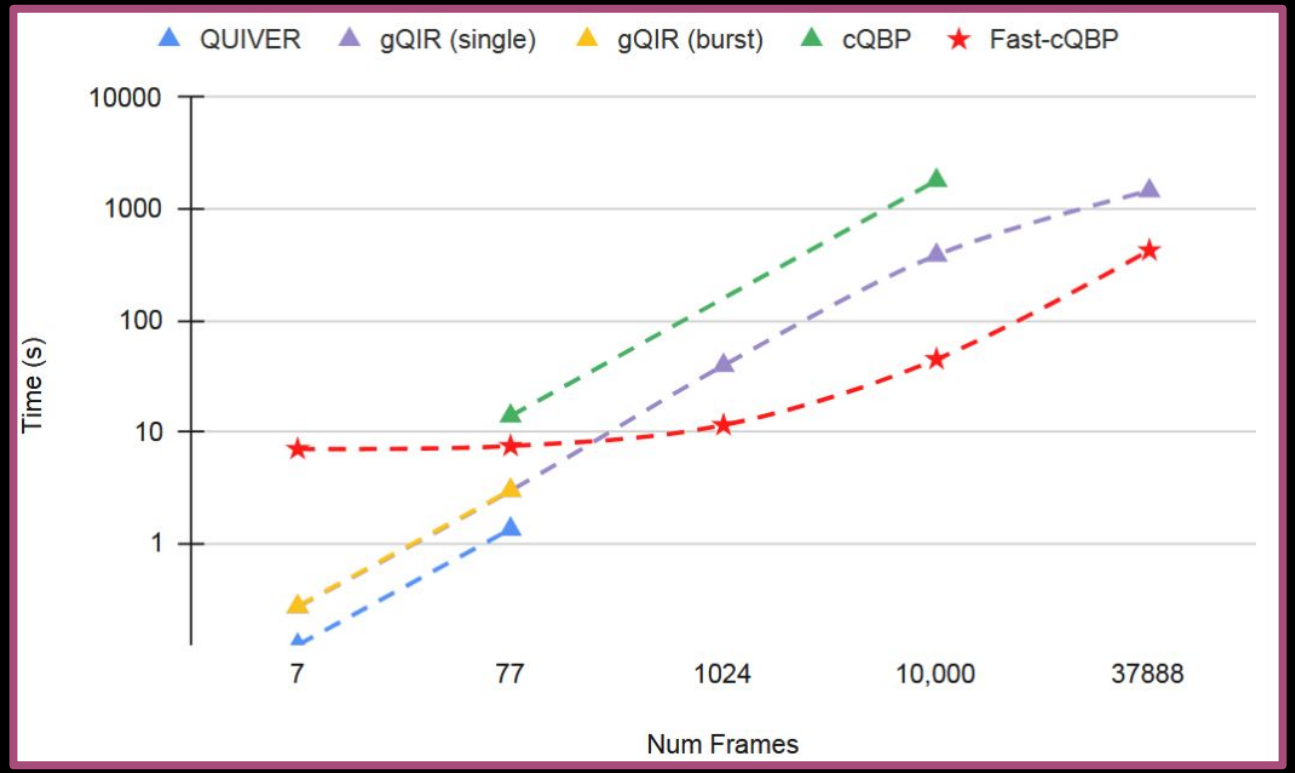
2. Compare against Generative Models (gQIR)
   a. Learn from data vs physics-based approach

3. Identify and demonstrate a key-weakness of largely pretrained generative methods as a stepping stone towards a generative HDR SPAD reconstruction solution

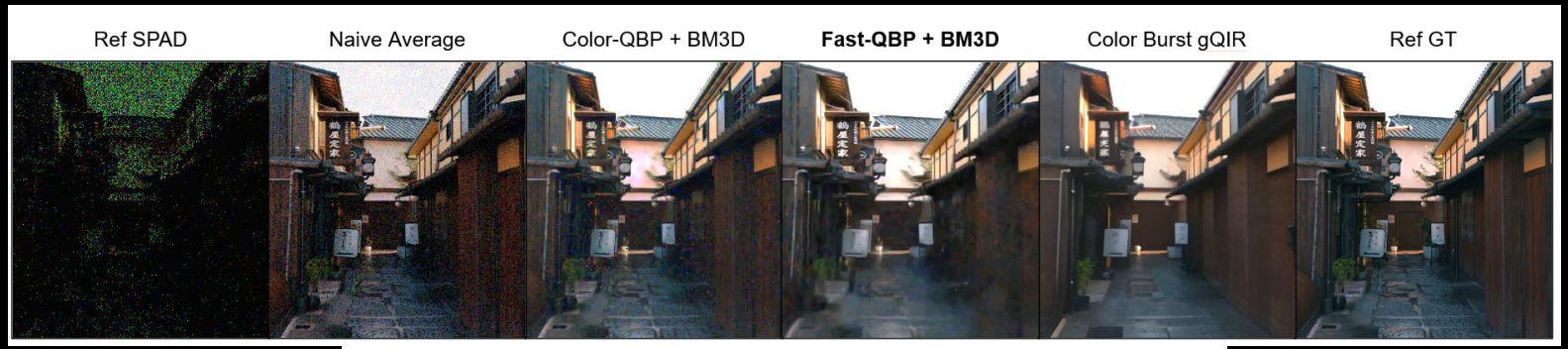
Fast cQBP Re-implementation: What Made it Faster?

1. Porting legacy MATLAB code to torch
2. Pinned memory for true async transfer
3. Separate CUDA stream for Parallelization
4. Concurrent data transfer and computation in all steps to hide latency:
    a. Align step
    b. Merge step
    c. Optional Super resolution step

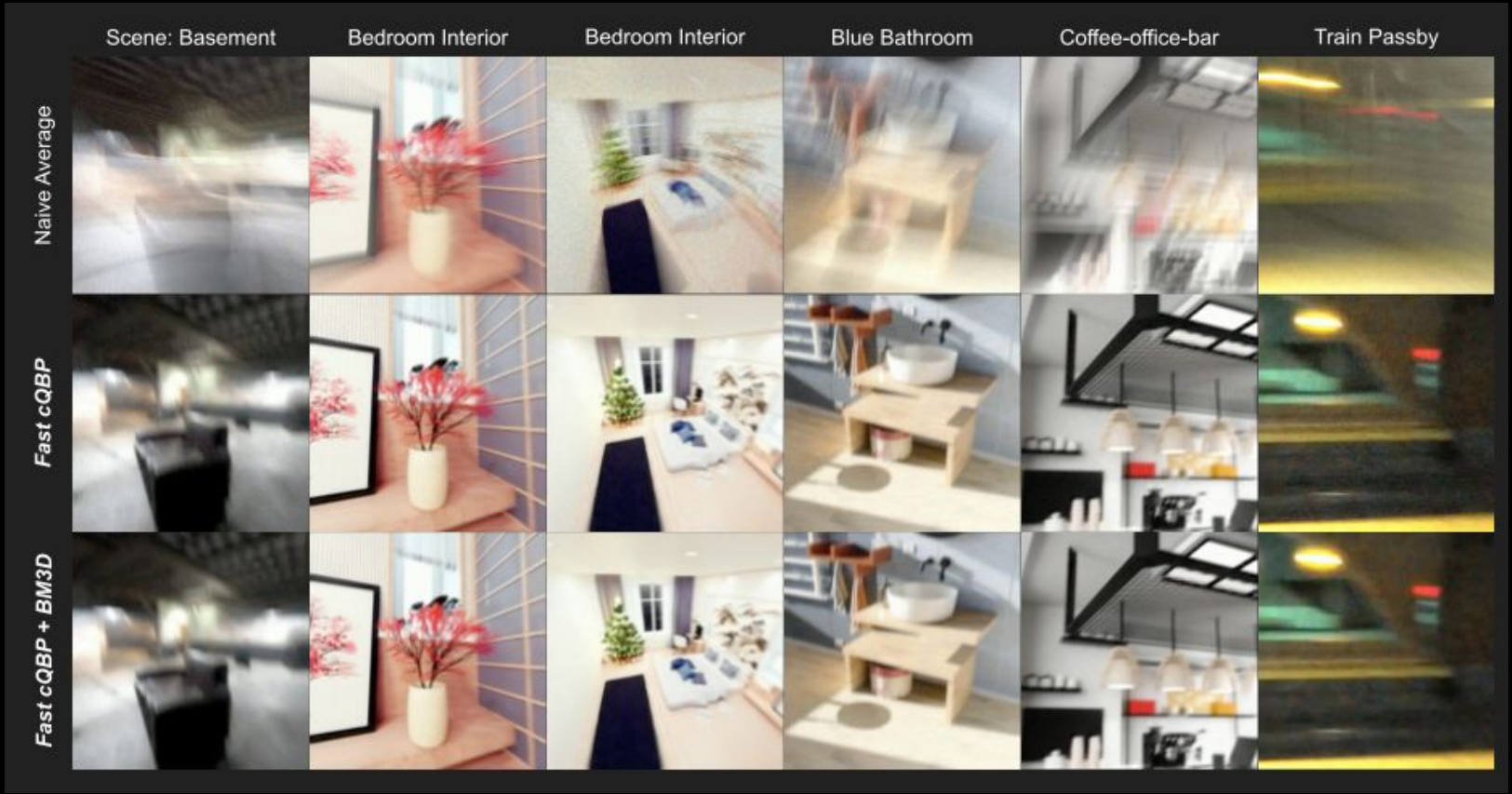
SpeedUp


QUIVER
gQIR (single)
gQIR (burst)
cQBP
Fast-cQBP
10000
1000
100
10
1
Time (s)
7
77
1024
10,000
37888
Num Frames

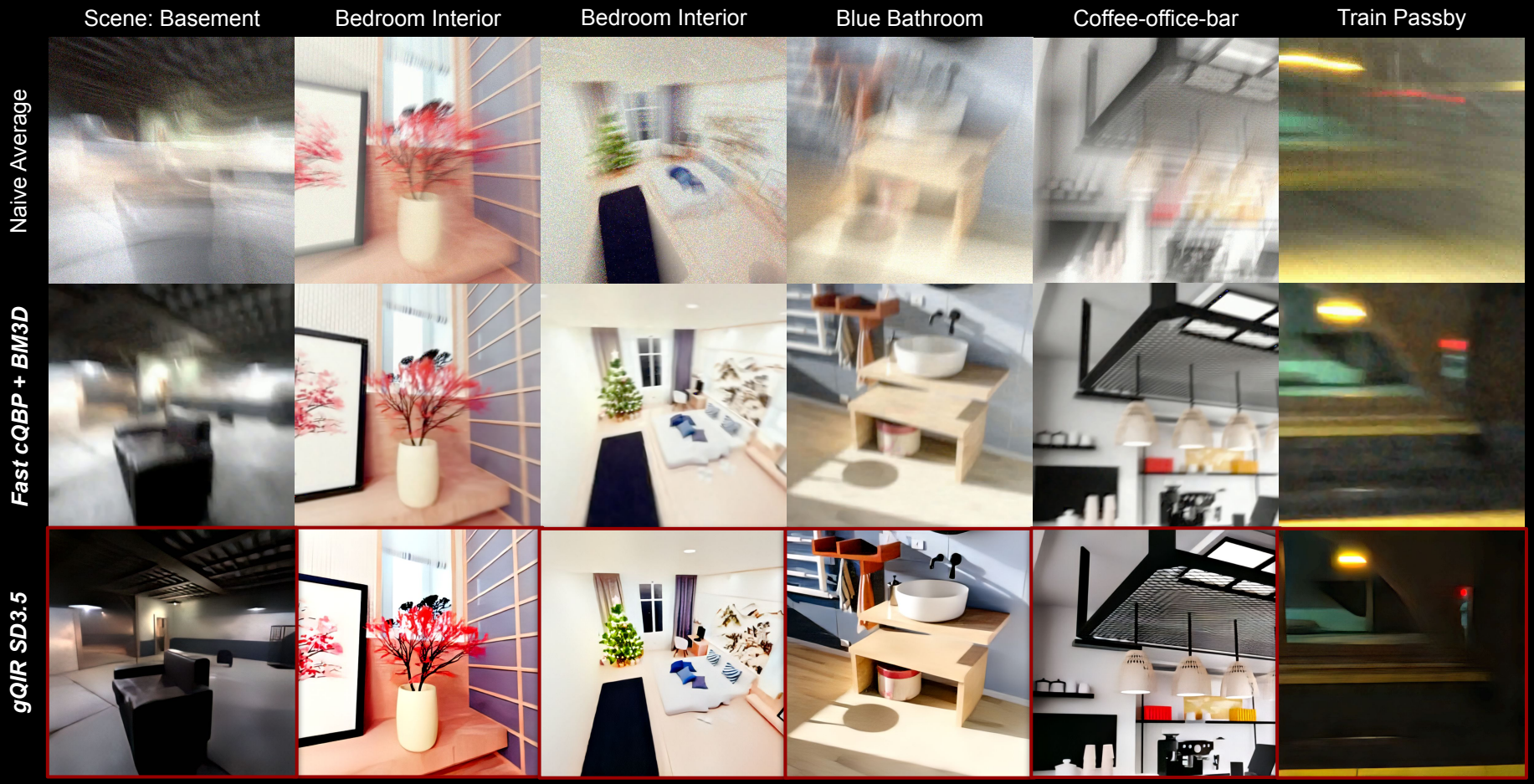
Faithfulness & Ablation Study - Sigma wrt Original



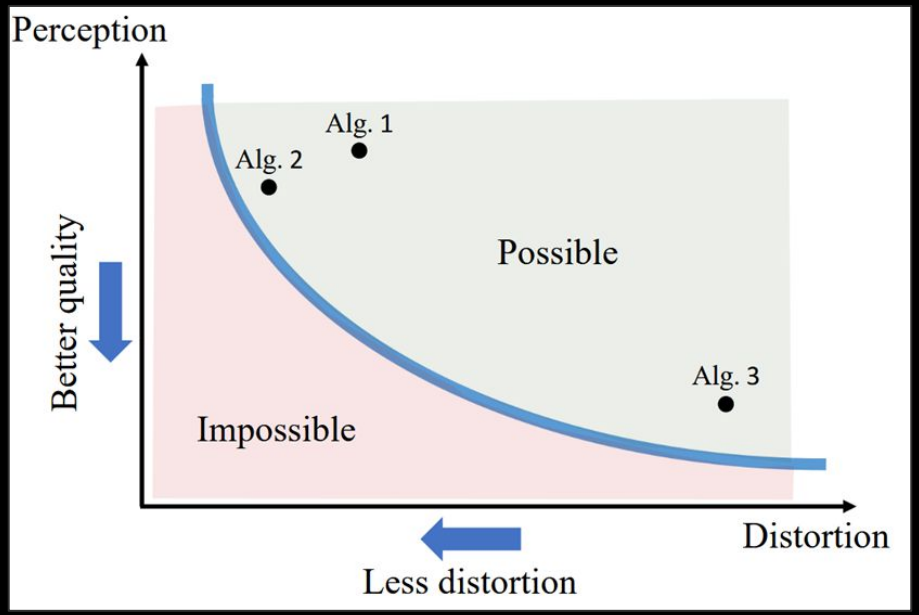
The Single Photon Challenge - Qualitative Results



Comparison with latest generative SPAD ISP: gQIR

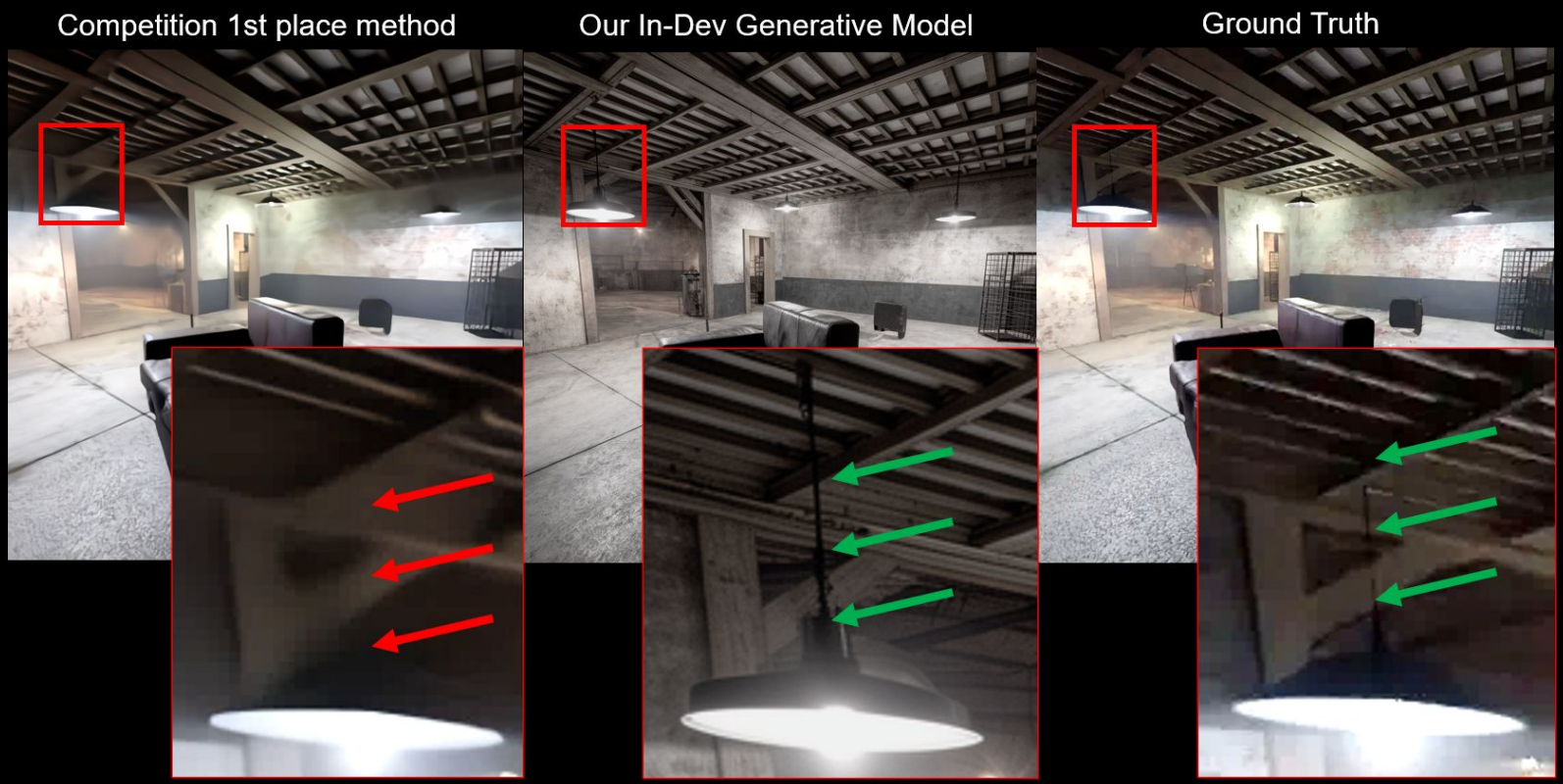


The Promise of Latching On To the Natural Image Manifold

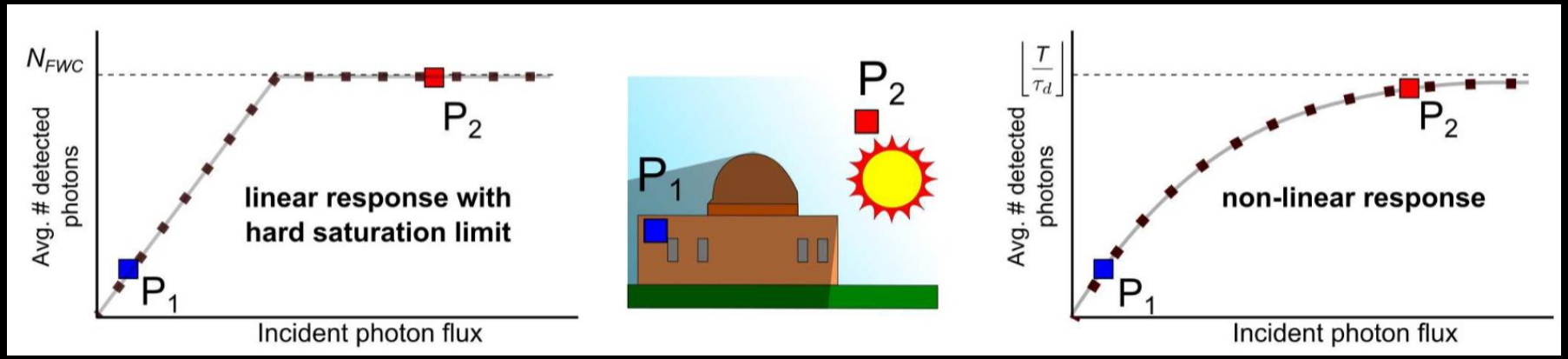


Supervised reconstruction is biased towards high distortion and does not respect the natural image manifold bounds!

Hanging Lamp in Basement Inset slide from PPT



What new did we learn?



But there’s a limitation that needs to be overcome first for generative models: HDR

DR Limitations in Image Generation due to LDR Pretraining

"Candid street portrait at night in Tokyo"

Problem:
DR Limitations in Image Generation due to LDR Pretraining

"Bright Living Room with Sunlight"

"Cat in a dark room"

"Rainy city street with neon lights"

Problem:
DR Limitations in Image Generation due to LDR Pretraining

Problem:
DR Limitations in Image Generation due to LDR Pretraining

Future Work:
Extending Generative Models to HDR

1. Fine-tune generative models using:
    a. Tonemapped HDR data
    b. Low-rank adaptation (LoRA-style)
2. Goal:
    a. Teach models physics-aware image formation